Motility associated genotypes of multiple *C. jejuni* isolates

Irene Ortega-Sanz[a], Beatriz Melero[a], Jordi Rovira[a]

[a] Department of Biotechnology and Food Science, University of Burgos, Burgos, Spain

Contact information: iortega@ubu.es

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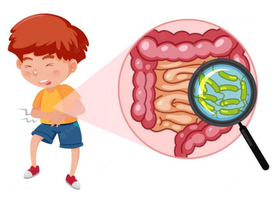
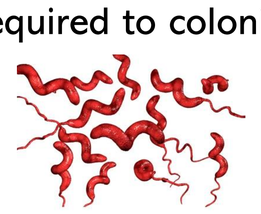
Background

Campylobacter jejuni is the leading cause of foodborne bacterial gastroenteritis worldwide that represents a potential threat to public health.

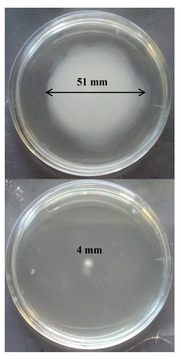


The ability of *C. jejuni* to survive and persist under harsh conditions is linked, among other factors, to flagella, that is required to colonize the gastrointestinal tract of humans and animals.



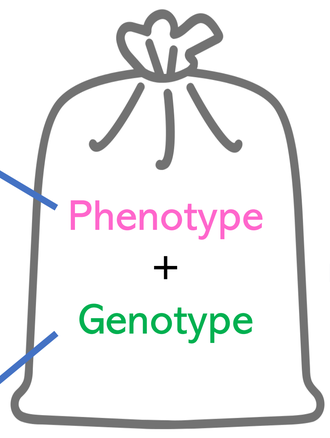
AIM examine the genetic basis for motility within 136 *C. jejuni* isolates

Methods



motility agar assay

- 0.4 % (w/v) Mueller-Hinton broth agar plate
- 5 µL of bacteria (~10^8 CFU/mL) stabbed in the centre of plate
- Incubation 37 °C 48 h
- Measure diameter of halo



INPUT

Low-motile < 3.5 cm

High-motile ≥ 3.5 cm

Scoary microbial pan-GWAS — binary phenotype + Pyseer — continuous phenotype

OUTPUT

gene presence/absence = PANGENOME

prokka
Roary

+

NCTC11168 vs **SNPs**

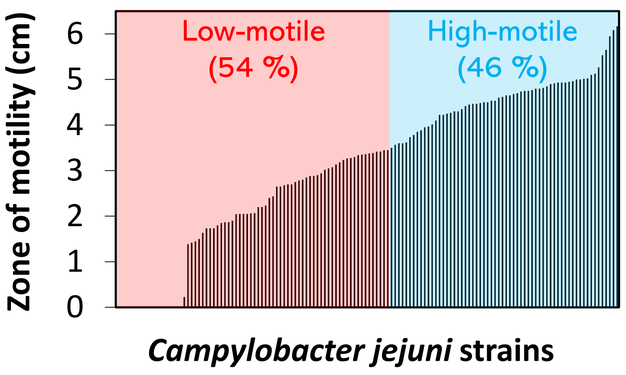
snippy
bcftools

Significant gen or SNP: $p < 0.05$

+

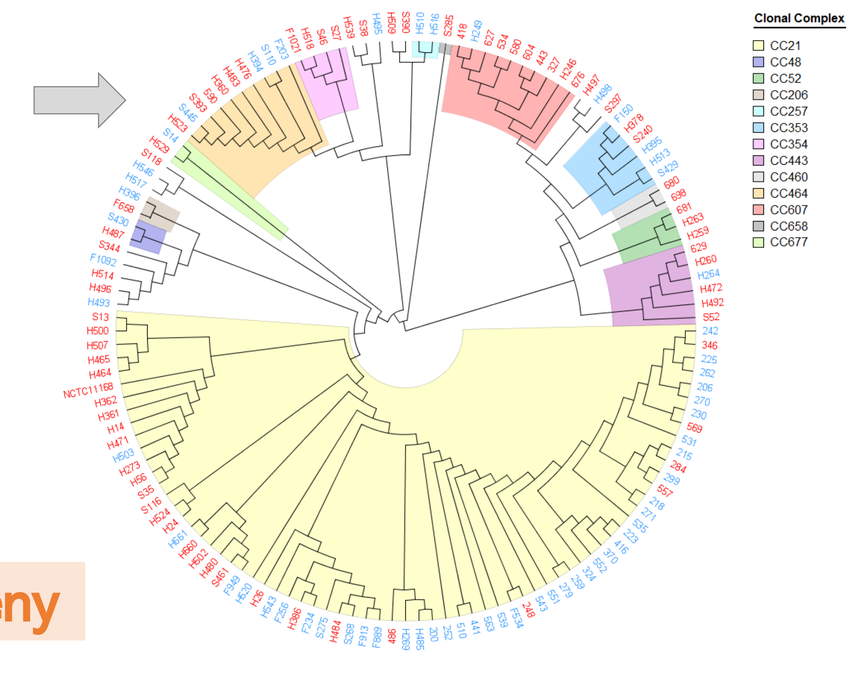
Gen or SNP present in both Scoary and Pyseer results

Results



Motility phenotype was widely distributed among phylogeny

Variation in swimming performance was observed for strains of the same lineage



- Genes significantly associated with motility in a 6,4 kb region containing genes coding for **transmembrane transporters**, **periplasmic** and **membrane proteins**. → gene synteny
- SNPs mostly linked to increased motility in clusters of genes (*cj0600-cj0601c* and *cj0803-cj0805*) coding for **transmembrane transporter proteins**, a **protease** and a **dehydrogenase** of the **membrane**.

Conclusions

- The structure and composition of membrane mediating nutrient transference might be important for energy availability for the flagella motor in *C. jejuni*.
- Unrevealing the genetic virulence mechanism of this pathogen is key to identify genetic markers for molecular-based risk assessment assay for human pathogenic *C. jejuni* or design of novel therapeutic targets.

FUTURE RESEARCH validation of candidate gens and SNPs in an independent cohort and/or further experimental

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Junta de Castilla y León